

```
C:\Program Files\MySQL\MySQL Server 8.0\bin>mysql -u root -p
```

```
Enter password: ****
```

```
Welcome to the MySQL monitor.  Commands end with ; or \g.
```

```
Your MySQL connection id is 11
```

```
Server version: 8.0.46 MySQL Community Server - GPL
```

```
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```

```
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owners.
```

```
Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.
```

```
mysql> show databases;
```

Database
bookflow_db
db2
employeedb1
information_schema
mysql
performance_schema
practice1
practice2
practice3
stored_procedures
studentdb1
sys

```
12 rows in set (0.00 sec)
```

```
mysql> use stored_procedures;
Database changed
mysql> CREATE TABLE IF NOT EXISTS employees (
->     emp_id INT PRIMARY KEY,
->     emp_name VARCHAR(50),
->     dept_id INT,
->     salary DECIMAL(10,2)
-> );
Query OK, 0 rows affected (0.02 sec)
```

```
mysql> ^C
mysql> INSERT INTO employees VALUES
-> (101, 'Rahul', 1, 50000),
-> (102, 'Priya', 2, 65000),
-> (103, 'Anil', 1, 55000),
-> (104, 'Sneha', 3, 70000),
-> (105, 'Kiran', 2, 48000);
Query OK, 5 rows affected (0.01 sec)
Records: 5  Duplicates: 0  Warnings: 0
```

```
mysql> select * from employees;
+-----+-----+-----+-----+
| emp_id | emp_name | dept_id | salary |
+-----+-----+-----+-----+
| 101 | Rahul | 1 | 50000.00 |
| 102 | Priya | 2 | 65000.00 |
| 103 | Anil | 1 | 55000.00 |
| 104 | Sneha | 3 | 70000.00 |
| 105 | Kiran | 2 | 48000.00 |
+-----+-----+-----+-----+
5 rows in set (0.00 sec)
```

```
mysql> DROP PROCEDURE IF EXISTS GetEmployees;  
Query OK, 0 rows affected, 1 warning (0.01 sec)
```

```
mysql> DELIMITER $$
```

```
mysql>
```

```
mysql> CREATE PROCEDURE GetEmployees()
```

```
-> BEGIN
```

```
->     SELECT * FROM employees;
```

```
-> END $$
```

```
Query OK, 0 rows affected (0.01 sec)
```

```
mysql>
```

```
mysql> DELIMITER ;
```

```
mysql>
```

```
mysql> CALL GetEmployees();
```

emp_id	emp_name	dept_id	salary
101	Rahul	1	50000.00
102	Priya	2	65000.00
103	Anil	1	55000.00
104	Sneha	3	70000.00
105	Kiran	2	48000.00

```
5 rows in set (0.00 sec)
```

```
Query OK, 0 rows affected (0.00 sec)
```

```
mysql> DROP PROCEDURE IF EXISTS GetEmployeeById;
```

```
Query OK, 0 rows affected, 1 warning (0.00 sec)
```

```
mysql>
mysql> DELIMITER ;
mysql>
mysql> CALL GetEmployeeById(102);
```

emp_id	emp_name	dept_id	salary
102	Priya	2	65000.00

1 row in set (0.00 sec)

Query OK, 0 rows affected (0.00 sec)

```
mysql> DROP PROCEDURE IF EXISTS IncreaseSalary;
Query OK, 0 rows affected, 1 warning (0.00 sec)
```

```
mysql> DELIMITER $$
```

```
mysql>
```

```
mysql> CREATE PROCEDURE IncreaseSalary(
->     IN eid INT,
->     IN amount DECIMAL(10,2)
-> )
-> BEGIN
->     UPDATE employees
->     SET salary = salary + amount
->     WHERE emp_id = eid;
-> END $$
```

Query OK, 0 rows affected (0.00 sec)

```
mysql>
```

```
mysql> DELIMITER ;
```

```
mysql>
```

```
mysql> CALL IncreaseSalary(101,5000);
```

Query OK, 1 row affected (0.00 sec)

```
mysql> DROP PROCEDURE IF EXISTS AvgSalary;
Query OK, 0 rows affected, 1 warning (0.00 sec)
```

```
mysql> DELIMITER $$
```

```
mysql>
```

```
mysql> CREATE PROCEDURE AvgSalary()
```

```
-> BEGIN
```

```
->     SELECT AVG(salary) AS AverageSalary
```

```
->     FROM employees;
```

```
-> END $$
```

```
Query OK, 0 rows affected (0.00 sec)
```

```
mysql>
```

```
mysql> DELIMITER ;
```

```
mysql>
```

```
mysql> CALL AvgSalary();
```

```
+-----+
```

```
| AverageSalary |
```

```
+-----+
```

```
| 58600.000000 |
```

```
+-----+
```

```
1 row in set (0.00 sec)
```

```
Query OK, 0 rows affected (0.00 sec)
```

```
mysql> CREATE TABLE IF NOT EXISTS departments (
```

```
->     dept_id INT PRIMARY KEY,
```

```
->     dept_name VARCHAR(50)
```

```
-> );
```

```
Query OK, 0 rows affected (0.02 sec)
```

```
mysql> INSERT IGNORE INTO departments VALUES
```

```
-> (1, 'CSE'), (2, 'ECE'), (3, 'EEE');
```

```
Query OK, 3 rows affected (0.00 sec)
```

```
Records: 3  Duplicates: 0  Warnings: 0
```

```
mysql>
mysql> DROP PROCEDURE IF EXISTS EmployeeDepartment;
Query OK, 0 rows affected, 1 warning (0.00 sec)

mysql> DELIMITER $$
mysql>
mysql> CREATE PROCEDURE EmployeeDepartment()
-> BEGIN
->     SELECT e.emp_name, d.dept_name, e.salary
->     FROM employees e
->     JOIN departments d ON e.dept_id=d.dept_id;
-> END $$
Query OK, 0 rows affected (0.00 sec)
```

```
mysql>
mysql> DELIMITER ;
mysql>
mysql> CALL EmployeeDepartment();
+-----+-----+-----+
| emp_name | dept_name | salary |
+-----+-----+-----+
| Rahul    | CSE       | 55000.00 |
| Priya    | ECE       | 65000.00 |
| Anil     | CSE       | 55000.00 |
| Sneha    | EEE       | 70000.00 |
| Kiran    | ECE       | 48000.00 |
+-----+-----+-----+
5 rows in set (0.00 sec)
```

```
Query OK, 0 rows affected (0.00 sec)
```

```
mysql> DROP PROCEDURE IF EXISTS DepartmentAverageSalary;  
Query OK, 0 rows affected, 1 warning (0.00 sec)
```

```
mysql> DELIMITER $$
```

```
mysql>
```

```
mysql> CREATE PROCEDURE DepartmentAverageSalary()  
-> BEGIN  
->     SELECT d.dept_name,  
->           COUNT(e.emp_id) AS EmployeeCount,  
->           AVG(e.salary) AS AverageSalary  
->     FROM employees e  
->     JOIN departments d ON e.dept_id=d.dept_id  
->     GROUP BY d.dept_id, d.dept_name;  
-> END $$
```

```
Query OK, 0 rows affected (0.00 sec)
```

```
mysql>
```

```
mysql> DELIMITER ;
```

```
mysql>
```

```
mysql> CALL DepartmentAverageSalary();
```

dept_name	EmployeeCount	AverageSalary
CSE	2	55000.000000
ECE	2	56500.000000
EEE	1	70000.000000

```
3 rows in set (0.00 sec)
```

```
mysql> DROP PROCEDURE IF EXISTS SalaryStatus;  
Query OK, 0 rows affected, 1 warning (0.00 sec)
```

```
mysql> DELIMITER $$
```

```
mysql>
```

```
mysql> CREATE PROCEDURE SalaryStatus(IN eid INT)  
-> BEGIN  
->     DECLARE sal DECIMAL(10,2);  
->  
->     SELECT salary INTO sal  
->     FROM employees  
->     WHERE emp_id=eid;  
->  
->     IF sal >= 60000 THEN  
->         SELECT 'High Salary' AS Status;  
->     ELSE  
->         SELECT 'Low Salary' AS Status;  
->     END IF;  
-> END $$
```

```
Query OK, 0 rows affected (0.00 sec)
```

```
mysql>
```

```
mysql> DELIMITER ;
```

```
mysql>
```

```
mysql> CALL SalaryStatus(102);
```

```
+-----+
```

```
| Status      |
```

```
+-----+
```

```
| High Salary |
```

```
+-----+
```

```
1 row in set (0.00 sec)
```

```
Query OK, 0 rows affected (0.00 sec)
```

```

mysql> DELIMITER $$
mysql>
mysql> CREATE PROCEDURE SalaryStatus(IN eid INT)
-> BEGIN
->     DECLARE sal DECIMAL(10,2);
->
->     SELECT salary INTO sal
->     FROM employees
->     WHERE emp_id=eid;
->
->     IF sal >= 60000 THEN
->         SELECT 'High Salary' AS Status;
->     ELSE
->         SELECT 'Low Salary' AS Status;
->     END IF;
-> END $$

```

Query OK, 0 rows affected (0.00 sec)

```

mysql>
mysql> DELIMITER ;
mysql>
mysql> CALL SalaryStatus(102);

```

```

+-----+
| Status      |
+-----+
| High Salary |
+-----+

```

1 row in set (0.00 sec)

Query OK, 0 rows affected (0.00 sec)

```

mysql> DROP PROCEDURE IF EXISTS EmployeeCount;
Query OK, 0 rows affected, 1 warning (0.00 sec)

```

```

mysql> DELIMITER $$
mysql>
mysql> CREATE PROCEDURE EmployeeCount(OUT total INT)
-> BEGIN
->     SELECT COUNT(*) INTO total

```

```
mysql> DELIMITER $$
mysql>
mysql> CREATE PROCEDURE EmployeeCount(OUT total INT)
-> BEGIN
->     SELECT COUNT(*) INTO total
->     FROM employees;
-> END $$
```

Query OK, 0 rows affected (0.00 sec)

```
mysql>
mysql> DELIMITER ;
mysql>
mysql> CALL EmployeeCount(@total);
Query OK, 1 row affected (0.00 sec)
```

```
mysql> SELECT @total AS TotalEmployees;
```

TotalEmployees
5

1 row in set (0.00 sec)

```
mysql> DROP PROCEDURE IF EXISTS AddBonus;
Query OK, 0 rows affected, 1 warning (0.00 sec)
```

```
mysql> DELIMITER $$
mysql>
mysql> CREATE PROCEDURE AddBonus(INOUT bonus DECIMAL(10,2))
-> BEGIN
->     SET bonus = bonus + 5000;
-> END $$
```

Query OK, 0 rows affected (0.01 sec)

```
mysql>
mysql> DELIMITER ;
mysql>
mysql> SET @bonus=10000;
Query OK, 0 rows affected (0.00 sec)
```

```

mysql> DELIMITER $$
mysql>
mysql> CREATE PROCEDURE TransferSalaryAmount(
->     IN fromEmp INT,
->     IN toEmp INT,
->     IN amount DECIMAL(10,2)
-> )
-> BEGIN
->     START TRANSACTION;
->
->     UPDATE employees
->     SET salary=salary-amount
->     WHERE emp_id=fromEmp;
->
->     UPDATE employees
->     SET salary=salary+amount
->     WHERE emp_id=toEmp;
->
->     COMMIT;
-> END $$

```

Query OK, 0 rows affected (0.00 sec)

```

mysql>
mysql> DELIMITER ;
mysql>
mysql> CALL TransferSalaryAmount(101,102,2000);
Query OK, 0 rows affected (0.00 sec)

```

```

mysql> SELECT * FROM employees WHERE emp_id IN (101,102);

```

emp_id	emp_name	dept_id	salary
101	Rahul	1	53000.00
102	Priya	2	67000.00

2 rows in set (0.00 sec)

2 rows in set (0.00 sec)

```
mysql> SHOW PROCEDURE STATUS
      -> WHERE Db='employeeedb1';
Empty set (0.01 sec)
```

```
mysql>
mysql> SHOW CREATE PROCEDURE GetEmployees;
+-----+-----+
+-----+-----+
+-----+-----+
| Procedure      | sql_mode
| Create Procedure
| Connection     | Database Collation |
+-----+-----+
+-----+-----+
+-----+-----+
| GetEmployees  | ONLY_FULL_GROUP_BY,STRICT_TRANS_TABLES,NO_ZERO_IN_DATE,NO_ZERO_DATE,ERROR_FOR
N | CREATE DEFINER='root'@'localhost' PROCEDURE `GetEmployees`()
BEGIN
    SELECT * FROM employees;
END | cp850          | cp850_general_ci      | utf8mb4_0900_ai_ci |
+-----+-----+
+-----+-----+
+-----+-----+
1 row in set (0.00 sec)
```

```
mysql>
mysql> DROP PROCEDURE IF EXISTS GetEmployees;
Query OK, 0 rows affected (0.00 sec)
```

```
mysql> DELIMITER $$
mysql> CREATE PROCEDURE Example()
      -> BEGIN
      ->     SELECT * FROM employees;
      -> END $$
```

```

mysql>
mysql> DROP PROCEDURE IF EXISTS GetEmployees;
Query OK, 0 rows affected (0.00 sec)

mysql> DELIMITER $$
mysql> CREATE PROCEDURE Example()
-> BEGIN
->     SELECT * FROM employees;
-> END $$
Query OK, 0 rows affected (0.01 sec)

mysql> DELIMITER ;
mysql> DROP PROCEDURE IF EXISTS GetHighSalaryEmployees;
Query OK, 0 rows affected, 1 warning (0.00 sec)

mysql> DELIMITER $$
mysql> CREATE PROCEDURE GetHighSalaryEmployees(IN minSalary DECIMAL(10,2))
-> BEGIN
->     SELECT *
->     FROM employees
->     WHERE salary > minSalary;
-> END $$
Query OK, 0 rows affected (0.00 sec)

mysql> DELIMITER ;
mysql>
mysql> CALL GetHighSalaryEmployees(55000);
+-----+-----+-----+-----+
| emp_id | emp_name | dept_id | salary  |
+-----+-----+-----+-----+
|    102 | Priya    |        2 | 67000.00 |
|    104 | Sneha    |        3 | 70000.00 |
+-----+-----+-----+-----+
2 rows in set (0.00 sec)

Query OK, 0 rows affected (0.00 sec)

mysql> DROP PROCEDURE IF EXISTS UpdateEmployeeDept;

```

```
mysql> DELIMITER $$
mysql> CREATE PROCEDURE UpdateEmployeeDept(IN eid INT, IN newDept INT)
-> BEGIN
->     UPDATE employees
->     SET dept_id = newDept
->     WHERE emp_id = eid;
-> END $$
```

Query OK, 0 rows affected (0.00 sec)

```
mysql> DELIMITER ;
mysql>
mysql> CALL UpdateEmployeeDept(105, 3);
Query OK, 1 row affected (0.00 sec)
```

```
mysql> DROP PROCEDURE IF EXISTS MaxSalary;
Query OK, 0 rows affected, 1 warning (0.00 sec)
```

```
mysql> DELIMITER $$
mysql> CREATE PROCEDURE MaxSalary(OUT maxSal DECIMAL(10,2))
-> BEGIN
->     SELECT MAX(salary) INTO maxSal
->     FROM employees;
-> END $$
```

Query OK, 0 rows affected (0.00 sec)

```
mysql> DELIMITER ;
mysql>
mysql> CALL MaxSalary(@maxSal);
Query OK, 1 row affected (0.00 sec)
```

```
mysql> SELECT @maxSal AS MaximumSalary;
```

MaximumSalary
70000.00

```
mysql> DROP PROCEDURE IF EXISTS DeptEmployeeCount;
Query OK, 0 rows affected, 1 warning (0.00 sec)
```

```
mysql> DELIMITER $$
mysql> CREATE PROCEDURE DeptEmployeeCount(IN did INT, OUT total INT)
-> BEGIN
->     SELECT COUNT(*) INTO total
->     FROM employees
->     WHERE dept_id = did;
-> END $$
Query OK, 0 rows affected (0.00 sec)
```

```
mysql> DELIMITER ;
mysql>
mysql> CALL DeptEmployeeCount(2, @total);
Query OK, 1 row affected (0.00 sec)
```

```
mysql> SELECT @total AS DeptEmployeeCount;
+-----+
| DeptEmployeeCount |
+-----+
|                  1 |
+-----+
1 row in set (0.00 sec)
```

```
mysql> DROP PROCEDURE IF EXISTS EmployeeDeptDetails;
Query OK, 0 rows affected, 1 warning (0.00 sec)
```

```
mysql> DELIMITER $$
mysql> CREATE PROCEDURE EmployeeDeptDetails()
-> BEGIN
->     SELECT e.emp_name, d.dept_name, e.salary
->     FROM employees e
->     JOIN departments d ON e.dept_id = d.dept_id;
-> END $$
Query OK, 0 rows affected (0.00 sec)
```

```
mysql> DELIMITER ;
mysql>
mysql> CALL EmployeeDeptDetails();
```

emp_name	dept_name	salary
Rahul	CSE	53000.00
Priya	ECE	67000.00
Anil	CSE	55000.00
Sneha	EEE	70000.00
Kiran	EEE	48000.00

```
5 rows in set (0.00 sec)
```

Query OK, 0 rows affected (0.01 sec)

```
mysql> DROP PROCEDURE IF EXISTS DeptHighAvgSalary;
Query OK, 0 rows affected, 1 warning (0.00 sec)
```

```
mysql> DELIMITER $$
mysql> CREATE PROCEDURE DeptHighAvgSalary()
-> BEGIN
->     SELECT d.dept_name, AVG(e.salary) AS AvgSalary
->     FROM employees e
->     JOIN departments d ON e.dept_id = d.dept_id
->     GROUP BY d.dept_id, d.dept_name
->     HAVING AVG(e.salary) > 55000;
-> END $$
```

Query OK, 0 rows affected (0.00 sec)

```
mysql> DELIMITER ;
mysql>
mysql> CALL DeptHighAvgSalary();
```

dept_name	AvgSalary
ECE	67000.000000
EEE	59000.000000

```
mysql> DROP PROCEDURE IF EXISTS SalaryCategory;
Query OK, 0 rows affected, 1 warning (0.00 sec)
```

```
mysql> DELIMITER $$
mysql> CREATE PROCEDURE SalaryCategory(IN eid INT)
-> BEGIN
->     DECLARE sal DECIMAL(10,2);
->     SELECT salary INTO sal FROM employees WHERE emp_id = eid;
->
->     IF sal >= 65000 THEN
->         SELECT 'High Salary' AS Category;
->     ELSEIF sal >= 50000 THEN
->         SELECT 'Medium Salary' AS Category;
->     ELSE
->         SELECT 'Low Salary' AS Category;
->     END IF;
-> END $$
Query OK, 0 rows affected (0.00 sec)
```

```
mysql> DELIMITER ;
mysql>
mysql> CALL SalaryCategory(105);
```

```
+-----+
| Category |
+-----+
| Low Salary |
+-----+
1 row in set (0.00 sec)
```

```
Query OK, 0 rows affected (0.00 sec)
```

```
mysql> DROP PROCEDURE IF EXISTS DeptAvgSalary;
Query OK, 0 rows affected, 1 warning (0.00 sec)
```

```
mysql> DELIMITER $$
mysql> CREATE PROCEDURE DeptAvgSalary(IN did INT, OUT avgSal DECIMAL(10,2))
-> BEGIN
```

```

mysql> DELIMITER ;
mysql>
mysql> CALL DeptAvgSalary(1, @avgSal);
Query OK, 1 row affected (0.00 sec)

mysql> SELECT @avgSal AS DeptAverageSalary;
+-----+
| DeptAverageSalary |
+-----+
|          54000.00 |
+-----+
1 row in set (0.00 sec)

mysql> DROP PROCEDURE IF EXISTS ApplyBonus;
Query OK, 0 rows affected, 1 warning (0.00 sec)

mysql> DELIMITER $$
mysql> CREATE PROCEDURE ApplyBonus(INOUT sal DECIMAL(10,2))
-> BEGIN
->     SET sal = sal + 5000;
-> END $$
Query OK, 0 rows affected (0.00 sec)

mysql> DELIMITER ;
mysql>
mysql> SET @sal = 45000;
Query OK, 0 rows affected (0.00 sec)

mysql> CALL ApplyBonus(@sal);
Query OK, 0 rows affected (0.00 sec)

mysql> SELECT @sal AS SalaryWithBonus;
+-----+
| SalaryWithBonus |
+-----+
|          50000.00 |
+-----+
1 row in set (0.00 sec)

mysql> DROP PROCEDURE IF EXISTS TransferDept;

```

```
mysql> SELECT @sal AS SalaryWithBonus;
```

```
+-----+  
| SalaryWithBonus |  
+-----+  
|          50000.00 |  
+-----+
```

```
1 row in set (0.00 sec)
```

```
mysql> DROP PROCEDURE IF EXISTS TransferDept;  
Query OK, 0 rows affected, 1 warning (0.00 sec)
```

```
mysql> DELIMITER $$
```

```
mysql> CREATE PROCEDURE TransferDept(IN eid1 INT, IN eid2 INT, IN newDept INT)  
-> BEGIN  
->     START TRANSACTION;  
->     UPDATE employees SET dept_id = newDept WHERE emp_id = eid1;  
->     UPDATE employees SET dept_id = newDept WHERE emp_id = eid2;  
->     COMMIT;  
-> END $$
```

```
Query OK, 0 rows affected (0.00 sec)
```

```
mysql> DELIMITER ;
```

```
mysql>
```

```
mysql> CALL TransferDept(101, 103, 2);
```

```
Query OK, 0 rows affected (0.00 sec)
```

```
mysql> |
```